

A Few Best Practices for Design and Implementation of Microservices

By embracing microservices and implementing the recommended best practices and tools, organisations can unlock agility, scalability, and innovation, paving the way for a successful digital transformation journey.



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Today's enterprises are undergoing digital transformation by adopting a variety of technologies. IoT, smart devices, analytics, mobility, and even social media are helping companies change their customer relationships, internal processes and value propositions. Microservices can accelerate this transformation, but implementing the right architecture may seem challenging. Fortunately, these potential barriers can be overcome by adopting several best practices.

Many organisations have key business functionalities fulfilled by a single, large 'monolithic' application — for example, a CRM.

By having functionality encapsulated in one application, the company's architecture is unable to provide the agility or scalability required to achieve digital transformation.

In contrast, a microservices architecture uses granular, distributed services with well-defined APIs based on domain-driven design. This structure promises improved agility, scalability, reliability, and a reduced time to market. Organisations that use microservices as a building block for their distributed enterprise systems can augment and aggregate existing functionality as a service, thus opening opportunities for new business models and revenue.

In addition, organisations must chalk-out a clearly defined foundation and design strategy for their microservice adoption. By following the best practices, they will be well on their way toward realising the many benefits microservices can offer and achieving their vision of digital transformation.

Drivers for microservices adoption

The main drivers for microservices architecture adoption by businesses are:

- The organisation's primary business revolves around extremely high volume transactions
- Rapid development requiring enhanced operational support
- The need to reduce implementation